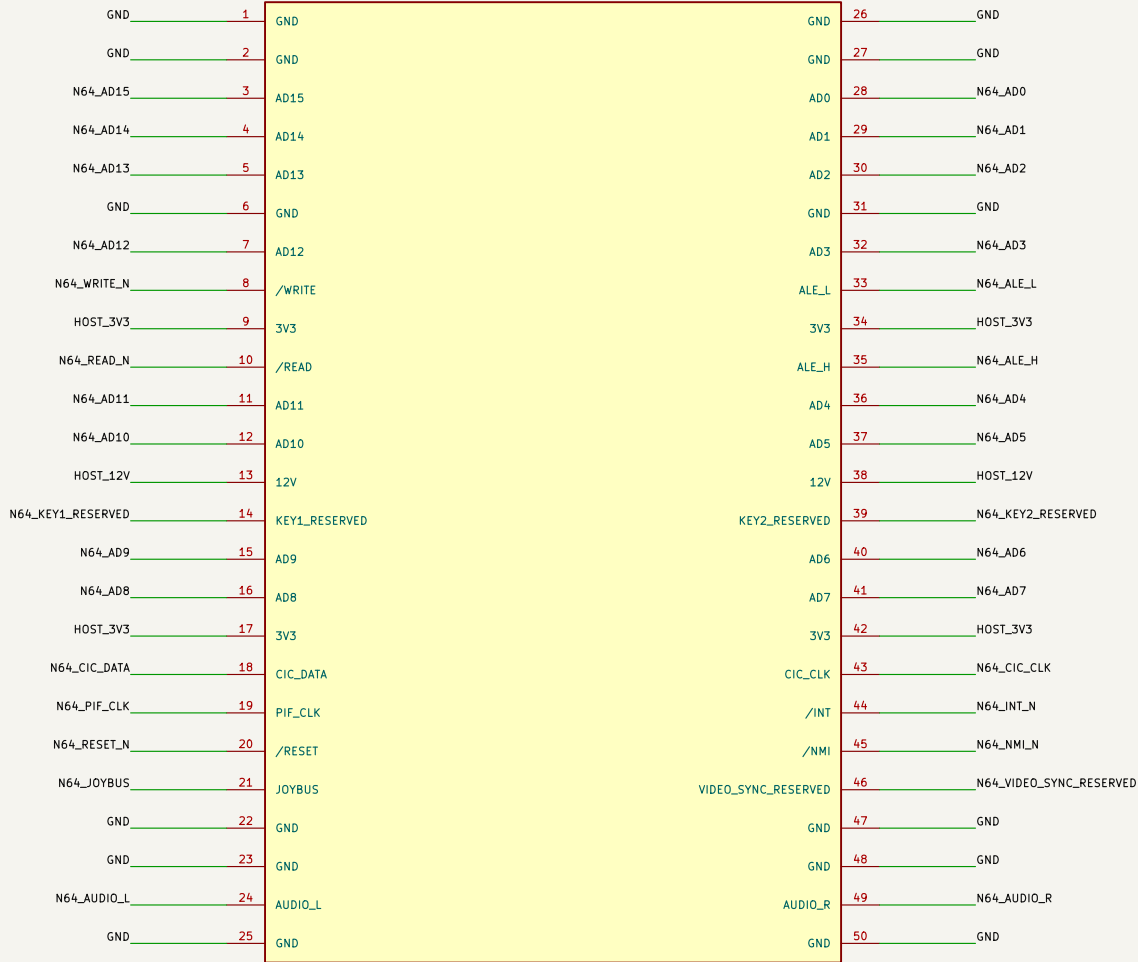


N64 / M64 HOST EDGE

Same 50-contact interface for both host systems

J1
N64_Cartridge_Edge_50



HOST INTERFACE
3.3 V logic; power inputs remain separate from local rails.
AD[15:0] is bidirectional. CIC_DATA / JOYBUS require release.
Reserved contacts and audio are identified only; no drivers fitted.
HOST_12V is reserved, with no selected load. No 5 V edge supply.

USB-C programmer

GND

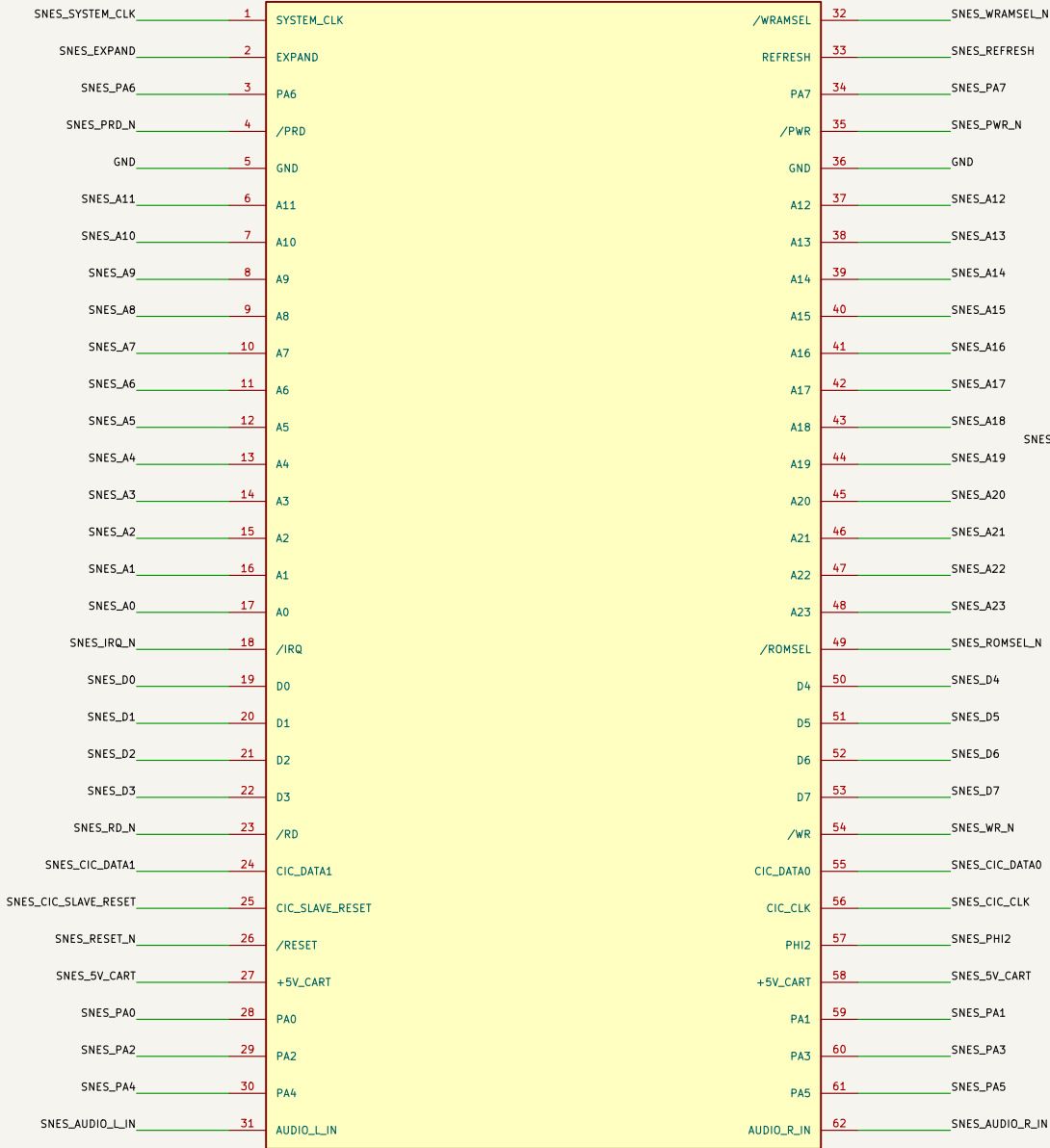
File: usb-programmer.kicad_sch

DRAFT BOUNDARY: cartridge connectors + USB programmer child sheet. Main FPGA, system power, cartridge translation and A/V remain pending.
Only GND is common between the two connectors. Matching names within one domain identify the same net; _N means active low.
Connector pin types are passive; electrical directions and source evidence are in interfaces/*.csv and docs/design/*-interface-notes.md.

SNES / SUPER FAMICOM SOCKET

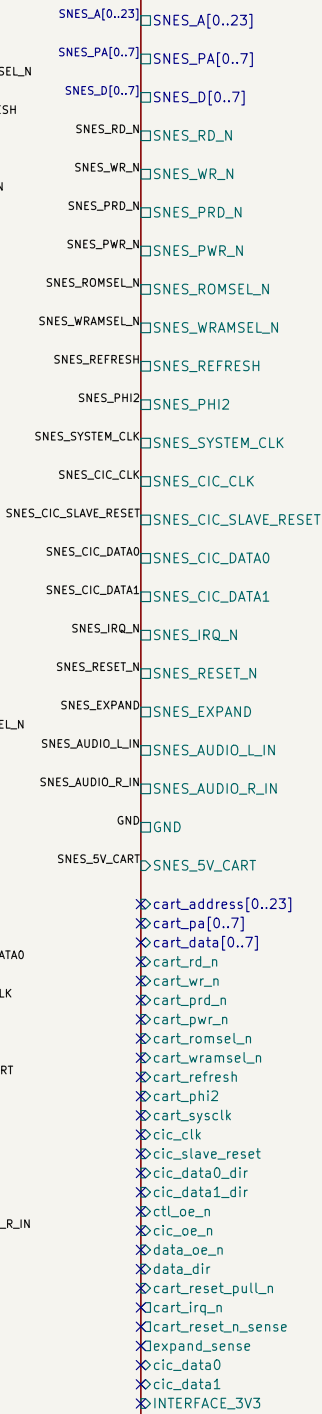
All 62 contacts retained, including analog cartridge audio

J2
SNES_Female_Slot_62



SNES INTERFACE
5 V cartridge logic requires reviewed translation and isolation.
SNES_5V_CART: protected switched supply, default OFF (pending).
System RESET must support cartridge assertion and input sensing.
CIC_SLAVE_RESET is separate; AUDIO_L/R_IN are analog inputs.
No universal NC contacts. Full pin behavior is recorded in the CSV.

SNES cartridge interface



File: cart-interface.kicad_sch

Source attribution and limits: hardware/sn64/README.md
INTERFACE DRAFT ONLY – NOT FOR MANUFACTURE

SN 64

Sheet: /
File: sn64.kicad_sch

Title: SN 64 – cartridge interfaces

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-28

Rev: 0.2-usb

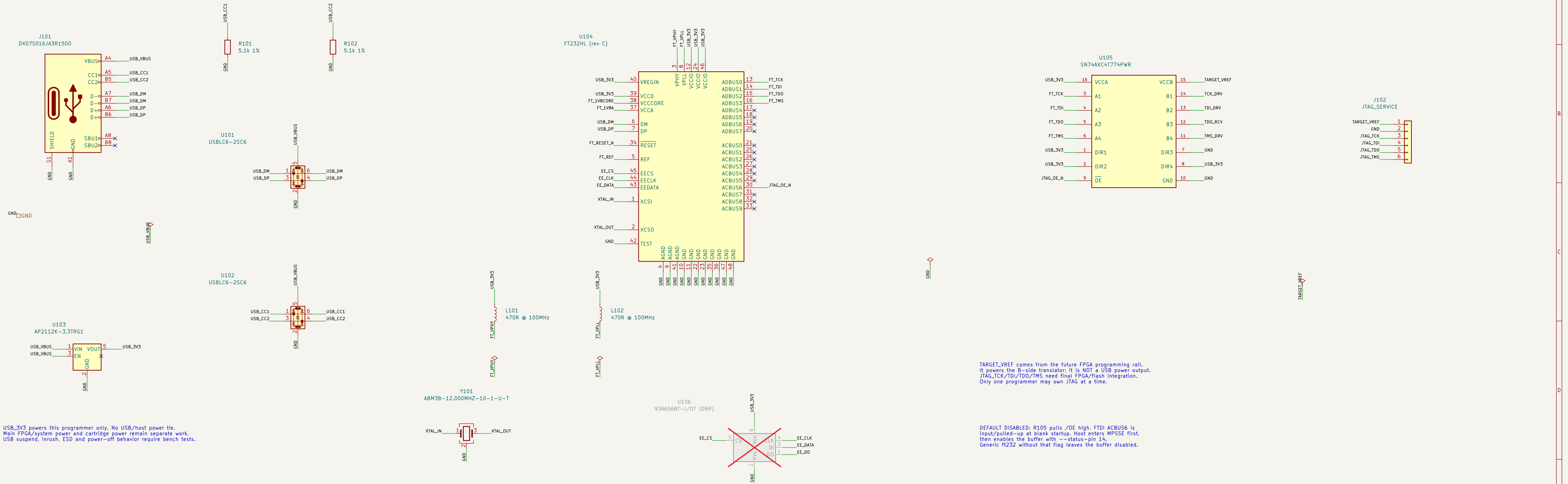
Id: 1/3

SIDE USB-C: INITIAL PROGRAMMING + RECOVERY

USB 2.0 device: 5V input only. Side location is mandatory: exact PCB/enclosure placement remains open.

FT232H HARDWARE BRIDGE

TARGET ISOLATION + SERVICE ACCESS

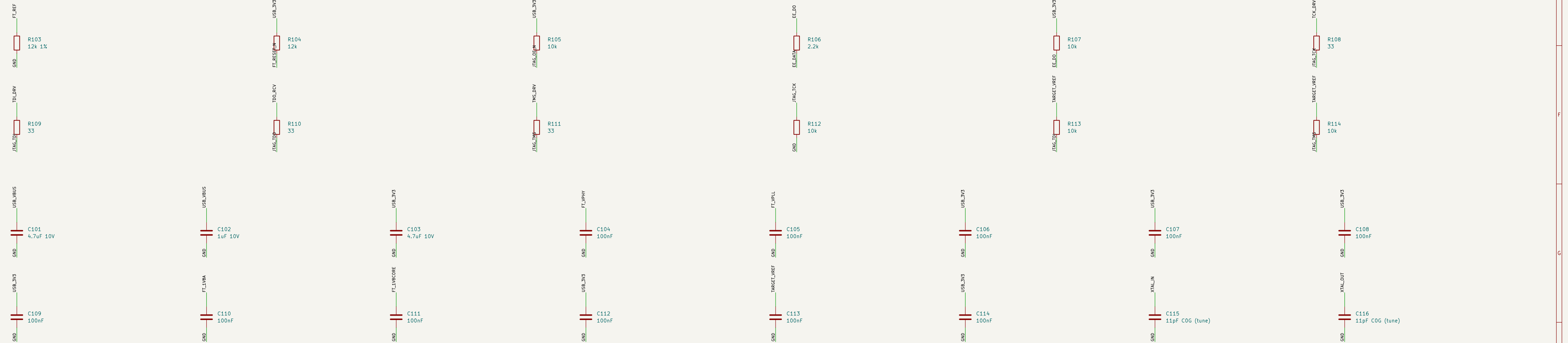


USB_3V3 powers this programmer only. No USB/host power tie. Main FPGA/system power and cartridge power remain separate work. USB suspend, inrush, ESD and power-off behavior require bench tests.

TARGET_VREF comes from the future FPGA programming rail. It powers the 8-side translator: it is NOT a USB power output. JTAG_TCK/TDI/TDO/TMS need final FPGA/flash integration. Only one programmer may own JTAG at a time.

DEFAULT DISABLED: R105 pulls /OE high. FTDI ACBUS6 is input/pulled-up at blank startup. Host enters MPSSE first, then enables the buffer with --status-pin 14. Generic ft232 without that flag leaves the buffer disabled.

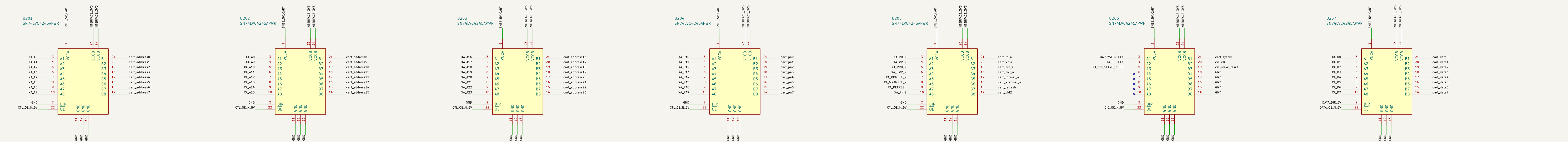
SUPPORT PASSIVES (place at their associated IC/pin; electrical grouping here is not PCB placement)



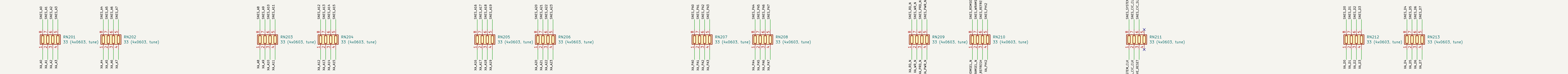
SNES CARTRIDGE –SIDE TRANSLATION AND PROTECTION

SN74LVC4245A: A port = 5 V VccA (VCCA = SNES_V5_CART), B port = 3.3 V VccB (VCCB = INTERFACE_V3V3); DIR/DE are referenced to VCCA (TI SCAS375K); DIR=L=H=A (FPGA drives socket); DIR=H=A=H; OE=H: Isolation.

U201–U206: console outputs, DIR strapped to GND (fixed FPGA→cartridge), U201–U205 share Ctl_De_N_V (bridge ctl_de_n); U206 (SYSTEM_CLK + CIC_CLK + CIC_SLAVE_RESET only) has its own Ctl_De_N_V (sn64_top_snes_cic_de_n, enabled from the sequencer IFACE state).
U207: DD–DT, DIR and DE both FPGA–controlled (bridge data_dir / data_de_s). Unused B inputs tied to GND; unused A outputs no-connect.

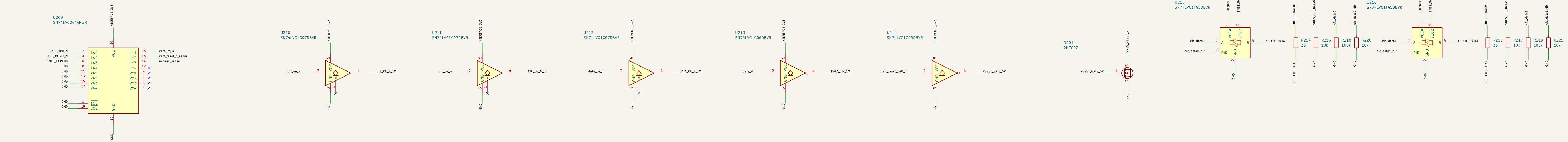


CARTRIDGE OUTPUT DAMPING: 33 R arrays at the translator A pins (source–series). Values are provisional; tune after SI measurement.



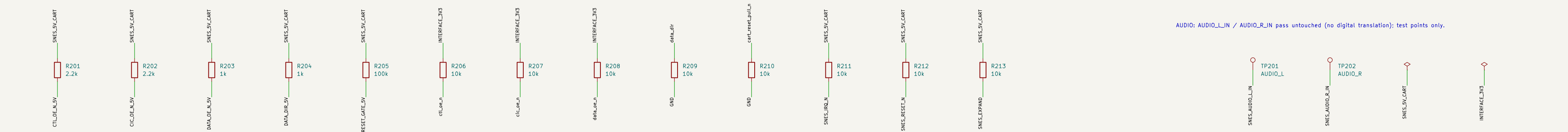
SENSE + RELEASE PATHS

U209 (SN74LVC244A) at 3.3 V, 5.5 V–tolerant inputs with buff. recovers /IRQ, /RESET, EXPAND (CIC data pin via U215/U216).
U210–U212 74LVC1G07 / U213 74LVC1G06, 3.3 V open–drain outputs (5.5 V tolerant, I_{OFF}) pulled to SNES_V5_CART.
No 5 V boot–up reaches an FPGA pin, or FPGA unconfigured → every translator /OE released high (isolated).
/RESET: Q201 open–drain sink, RESET_GATE_V3V3 pulled to 5 V; cartridge held in reset unless INTERFACE_V3V3 is up AND the FPGA drives cart_reset_pull_n=1.

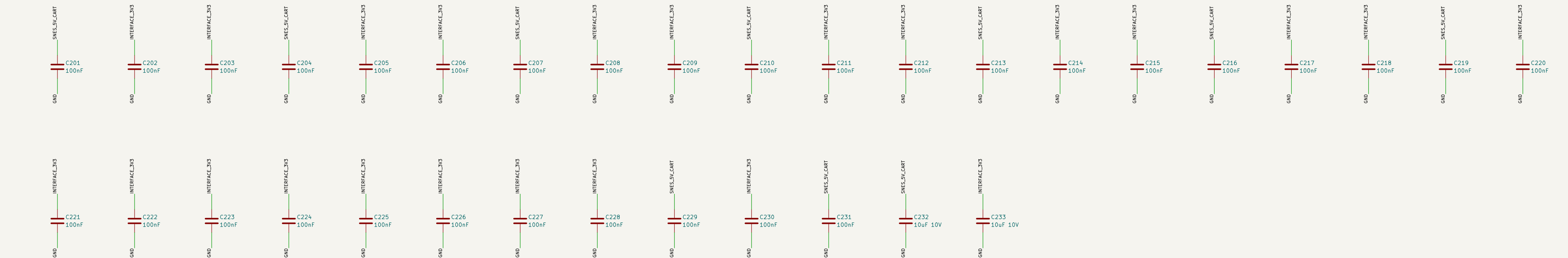


/OE and DIR edge timing: the bridge releases the data octet one A6.56 ns clock before turnaround. R203/R204 = 1k keep the passive rising edge short (about 10 pF load; tau < 10 ns). R201/R202 2.2k drive fly/one /OE inputs where only permit–loss timing matters. Verify on hardware.

PULL/BIAS: /IRQ, /RESET, EXPAND 10k to cartridge 5 V (OpenSFC keeps pull–ups on IRQ/EXPAND); CIC_DATA0 1 10k pull–downs (R216/R217). Values provisional.



DECOUPLING: one 100 nF per supply pin (place at the pin) plus one 10 μF bulk per rail.



DEMT 0.51–cart: schematic circuit candidate only. No PCB, SI simulation, power sequencing test, ESD part or cartridge test has been performed.
SNES_V5_CART and INTERFACE_V3V3 are hierarchical inputs from the (future) power sheet: #U201/#U202 main them externally supplied for ERC only.
Open socket ESD array selection, final damping/pull values: CIC pull–downs value to confirm against a real key CIC.
Source: TI SCAS375K (SN74LVC4245A), SCAS374AG (SN74LVC244A), SCE5296AG (SN74LVC1G07), SCE5296AB (SN74LVC1G06), SCE5515N (SN74LVC145).
See docx/design/cart–interface–schematic.mxd.

HIERARCHICAL PORTS – CARTRIDGE SIDE (root J2 net labels)

SNES_A[0..23]→SNES_A[0..23]
SNES_PA[0..7]→SNES_PA[0..7]
SNES_DI[0..7]→SNES_DI[0..7]
SNES_RD_N→SNES_RD_N
SNES_WR_N→SNES_WR_N
SNES_PRO_N→SNES_PRO_N
SNES_PWR_N→SNES_PWR_N
SNES_ROMSEL_N→SNES_ROMSEL_N
SNES_WRAMSEL_N→SNES_WRAMSEL_N
SNES_REFRESH→SNES_REFRESH
SNES_PHI2→SNES_PHI2
SNES_SYSTEM_CLK→SNES_SYSTEM_CLK
SNES_CIC_CLK→SNES_CIC_CLK
SNES_CIC_SLAVE_RESET→SNES_CIC_SLAVE_RESET
SNES_CIC_DATA0→SNES_CIC_DATA0
SNES_CIC_DATA1→SNES_CIC_DATA1
SNES_IRQ_N→SNES_IRQ_N
SNES_RESET_N→SNES_RESET_N
SNES_EXPAND→SNES_EXPAND
SNES_AUDIO_L_IN→SNES_AUDIO_L_IN
SNES_AUDIO_R_IN→SNES_AUDIO_R_IN
GND→GND
SNES_V5_CART→SNES_V5_CART

PENDING PORTS – FPGA SIDE (bridge port names) = DEFAULT–OFF RAILS FROM FUTURE POWER SHEET

cart_address[0..23]→cart_address[0..23]
cart_pa[0..7]→cart_pa[0..7]
cart_data[0..7]→cart_data[0..7]
cart_de_n→cart_de_n
cart_wr_n→cart_wr_n
cart_rd_n→cart_rd_n
cart_prd_n→cart_prd_n
cart_pwr_n→cart_pwr_n
cart_romsel_n→cart_romsel_n
cart_wramsel_n→cart_wramsel_n
cart_refresh→cart_refresh
cart_phi2→cart_phi2
cart_sysclk→cart_sysclk
cic_clk→cic_clk
cic_slave_reset→cic_slave_reset
cic_data0_dir→cic_data0_dir
cic_data1_dir→cic_data1_dir
ctl_de_n→ctl_de_n
cic_or_n→cic_or_n
data_or_n→data_or_n
data_dir→data_dir
cart_reset_pull_n→cart_reset_pull_n
cart_irq_n→cart_irq_n
cart_reset_n_sense→cart_reset_n_sense
expand_sense→expand_sense
cic_data0→cic_data0
cic_data1→cic_data1
INTERFACE_V3V3→INTERFACE_V3V3

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Shiren /SNES cartridge interface/
File: cart–interface.kicad_sch

Title: SN 64 – SNES cartridge interface

Size: A1 Date: 2026–09–29

Kicad E.D.A. 10.0.6

Rev: 0.31–cart

16 / 33